



《CAD与工程制图》

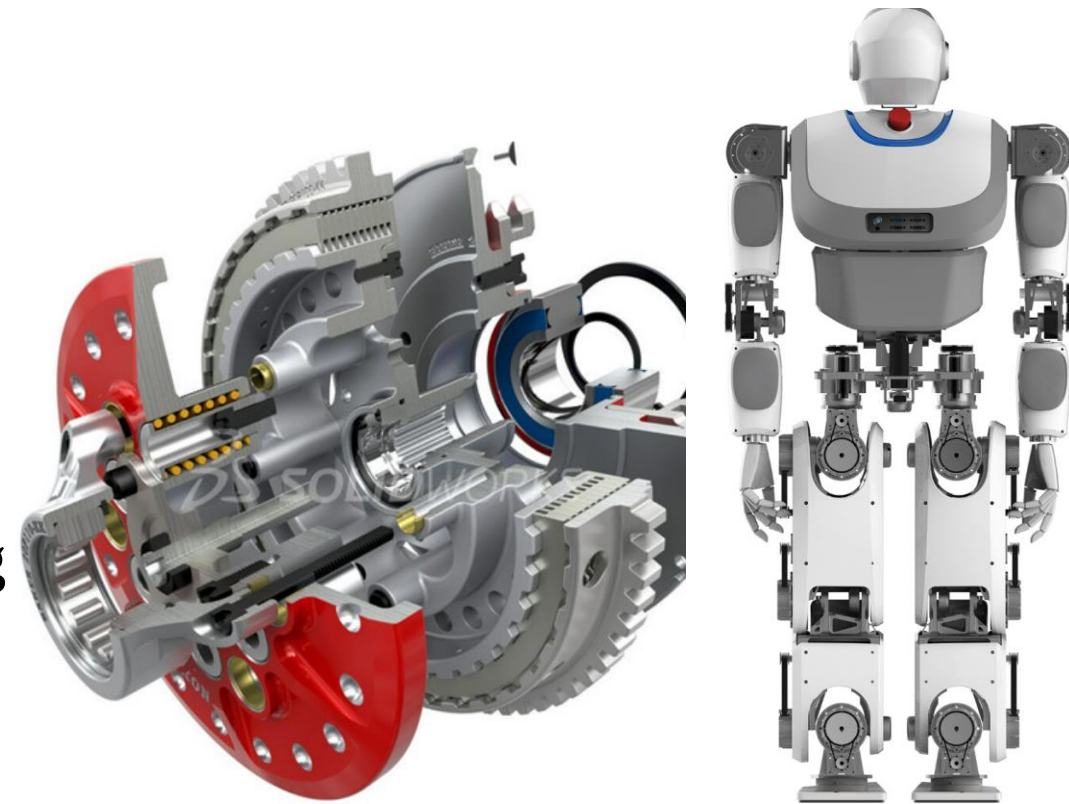
CAD (Computer Aided Design) and Engineering Drawing

2024年 秋季

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Southern University of Science and Technology



Chapter 4: Basic Bodies and Their Intersection Lines

- **4-1 Basic Planar Solids**
- **4-2 Intersection Lines of Basic Planar Solids**
- **4-3 Basic Rotational Solids**
- **4-4 Intersection Lines of Rotational Solids**
- **4-5 Dimensioning of Basic Solids**

Learning Objectives

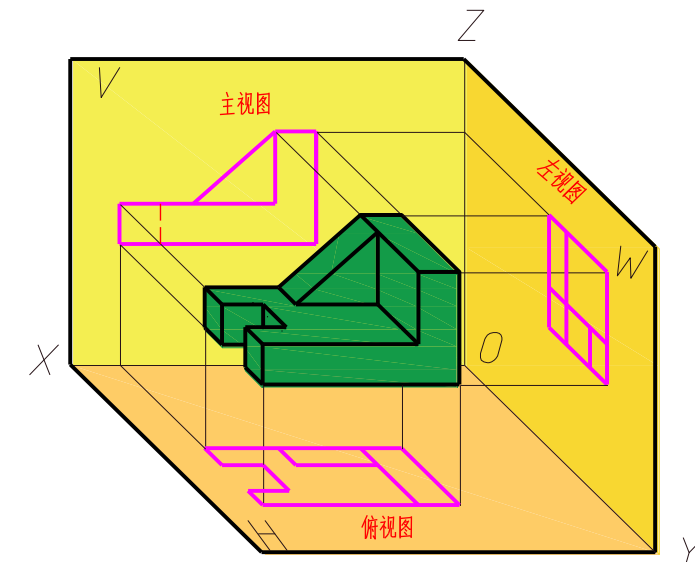
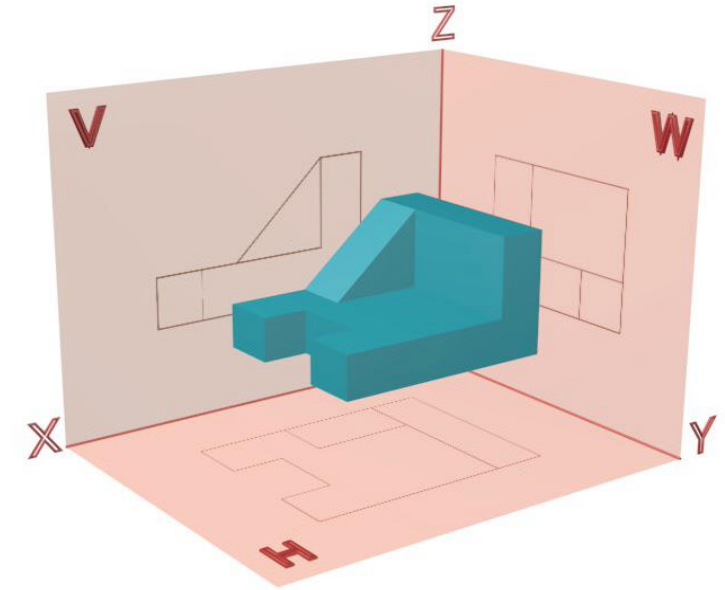
- **Familiarize with the Projections of Basic Solids**
- **Master the Projection and Drawing Methods of Intersection Lines**

1、 Formation of the Three-View Projection

The graphical representation obtained by projecting an object onto the projection plane is called a **view**.

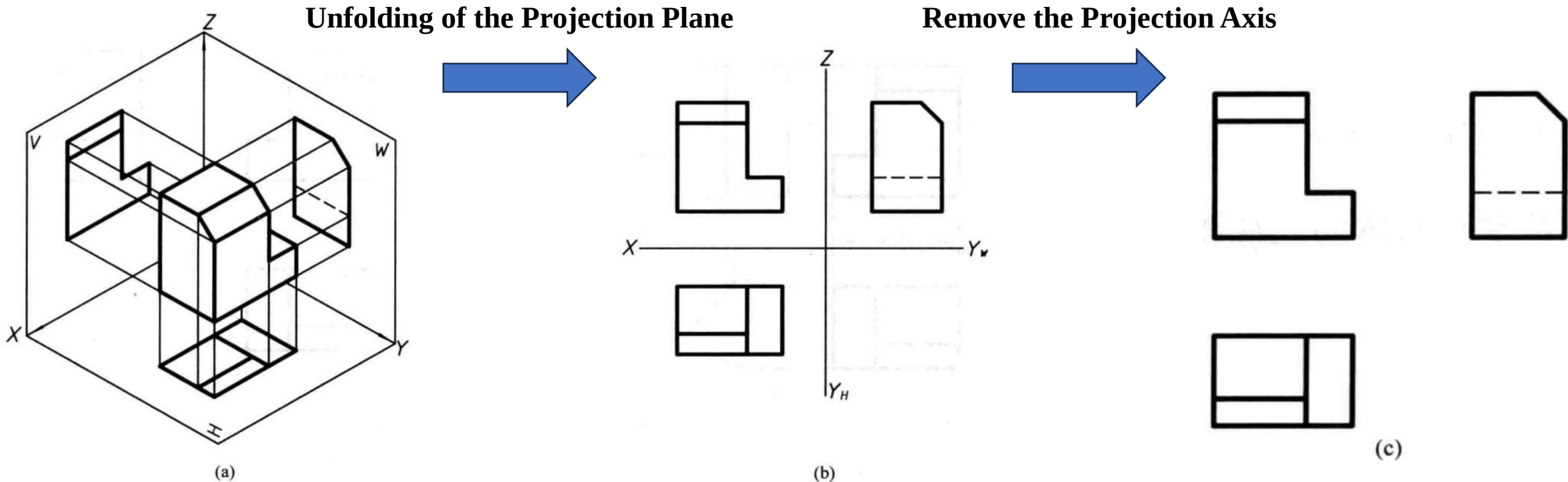
The projections of the object in the three projection plane system are as follows:

- V-plane projection — **Front view**
- H-plane projection — **Top view**
- W-plane projection — **Left side view**



1、 Formation of the Three-View Projection

Front view — Frontal projection Top view — Horizontal projection Left side view — Lateral projection

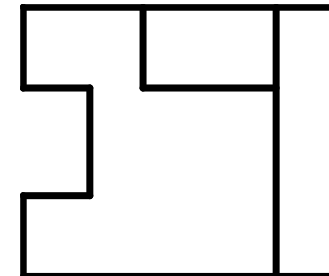
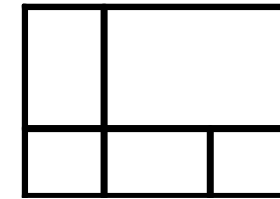
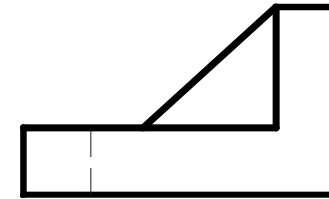


2、 The Relationship Between the Three Views

(1) Positional Relationship

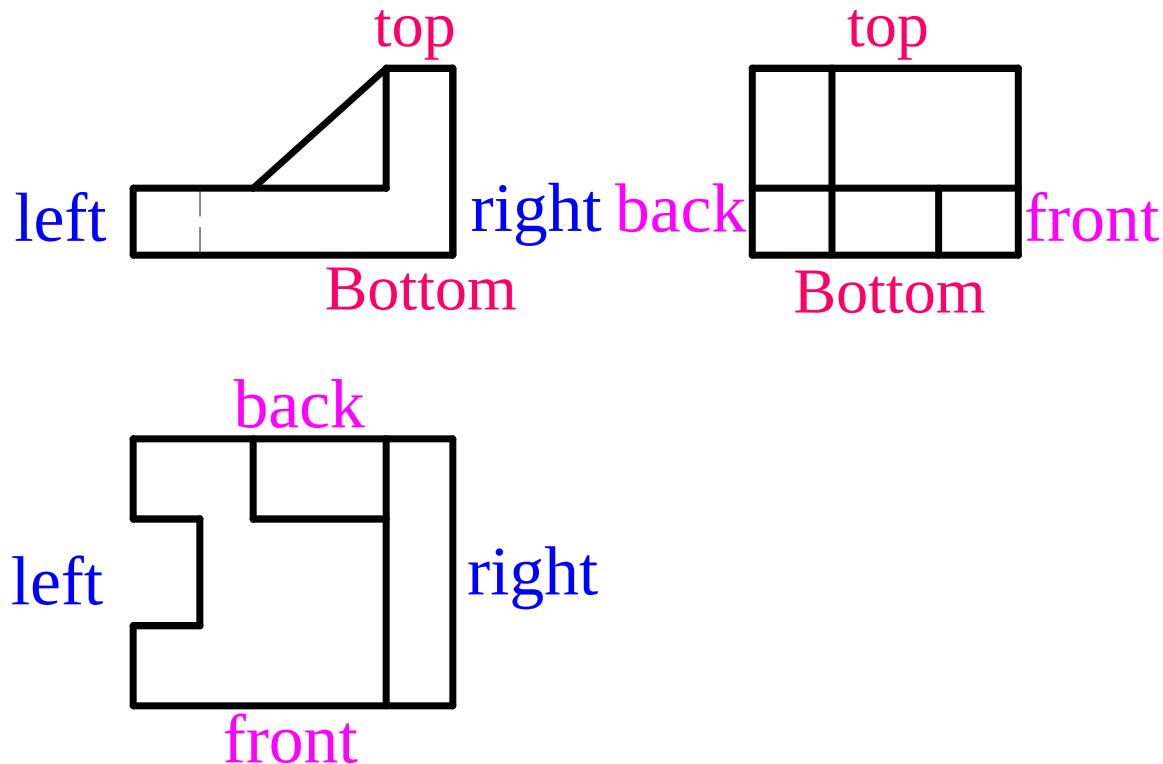
With the front view as the reference:

- The top view is directly below the front view.
- The left side view is directly to the right of the front view.



2、 The Relationship Between the Three Views

(2) Orientation Relationship



Front view: Reflects the object's top, bottom, and left, right.

Top view: Reflects the object's front, back, and left, right.

Left side view: Reflects the object's top, bottom, and front, back.

2、 The Relationship Between the Three Views

(3) Measurement Relationship

From left to right represents **the length of the object.**

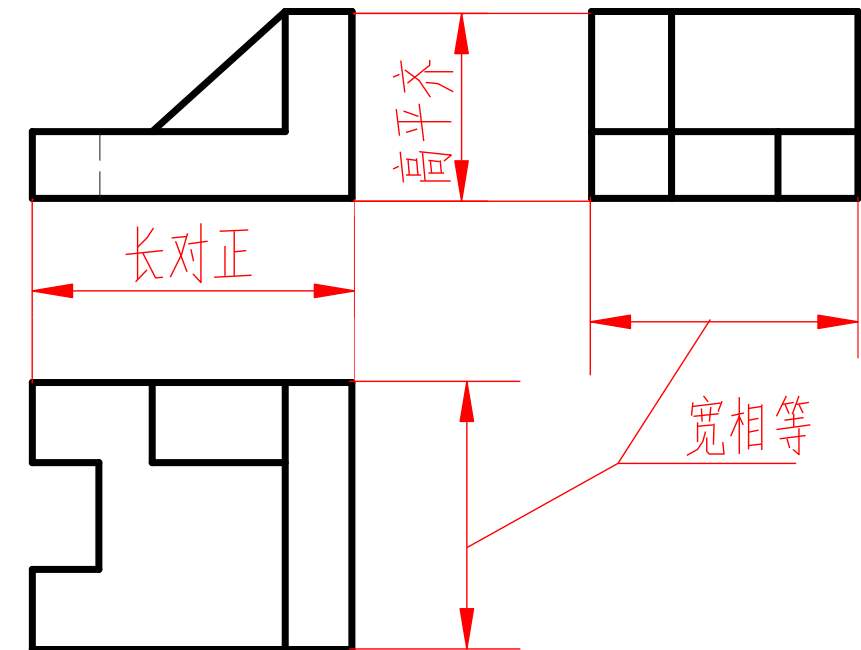
From front to back represents **the width of the object.**

From bottom to top represents **the height of the object.**

The length in the front and top views aligns.

The height in the front and left side views is equal.

The width in the top and left side views is the same.



Chapter 4: Basic Bodies and Their Intersection Lines

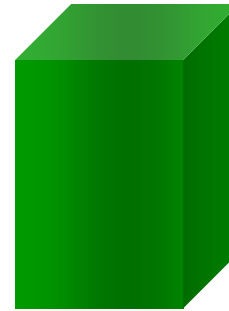
- **4-1 Basic Planar Solids**
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Classification of Basic Solids

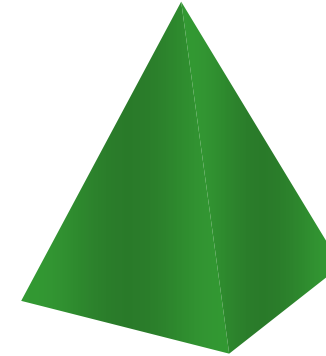
Common
Basic
Solids

Planar Solids

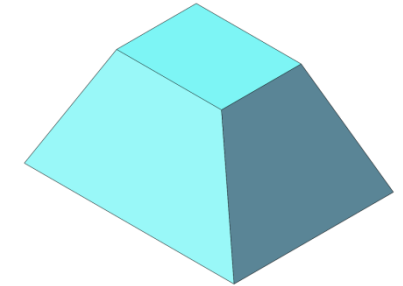
Solids whose surfaces are entirely enclosed by planes.



Rectangular Prism
(Four-sided Prism)



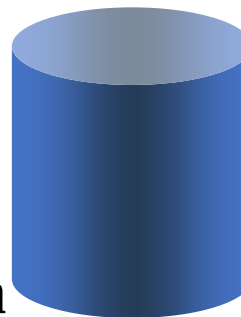
Triangular
Pyramid



Frustum of a
Square Pyramid

Curved Surface Solid

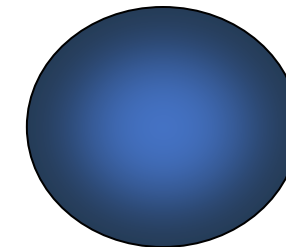
A solid whose surfaces are composed of curved surfaces or a combination of curved and planar surfaces.



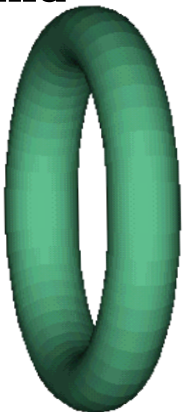
Cylinder



Cone



Sphere



Annulus
(Ring)

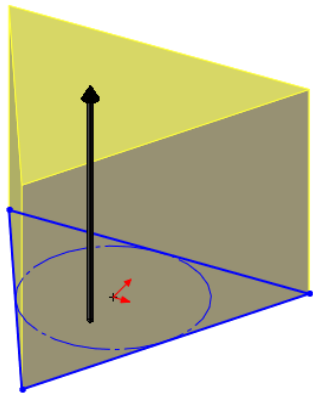
Review: 3-3 Projection of a Surface



	Perpendicular Plane	Vertical Plane	Lateral Plane
Solid View			
Projection View			
Faces on the Solid			

1、 Formation of a Prism

A prism can be considered as formed by stretching a planar shape, with the stretching direction perpendicular to the plane shape (cross-section).



**Triangular
Prism**



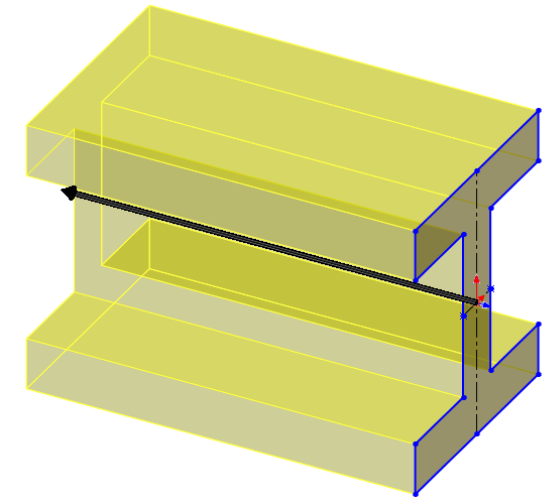
**Hexagonal
Prism**



**Four-sided
Prism**



**Concave
Prism**



I-beam Prism

1、Prism

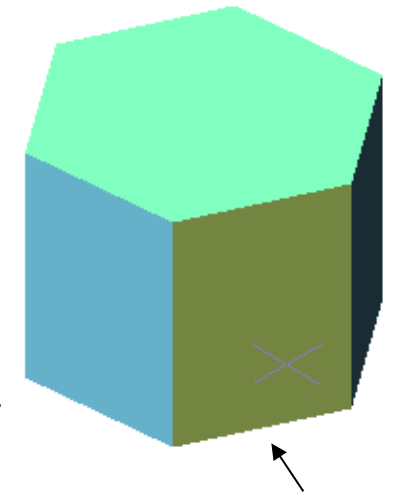
Drawing Steps:

- **Align** **How to choose the projection direction:** Position the solid so that the primary surfaces and edges become special position planes and special position lines on the projection plane.

- **Analysis** **Surface composition:** Top and bottom faces, six rectangular faces.

Projection characteristics:

- The top and bottom faces are horizontal planes, and the horizontal projection reflects the true shape, which is a regular hexagon. The front and side projections converge to a single straight line.
- The front and rear rectangular faces are **frontal plane**, and based on their spatial relative positions, the front projection reflects the true shape, while the horizontal and side projections converge to a straight line.
- The four lateral rectangular faces are **vertical planes**. Based on their spatial relative positions, the horizontal projection converges to a straight line, while the front and side projections are similar in shape.

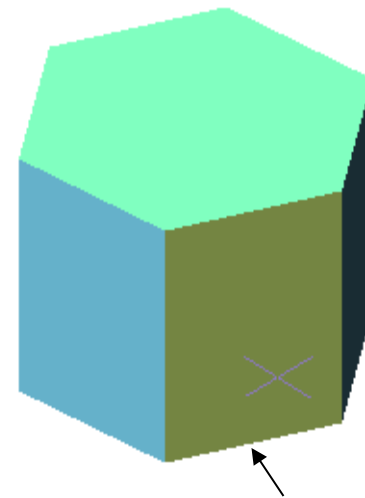


Front Projection
Direction

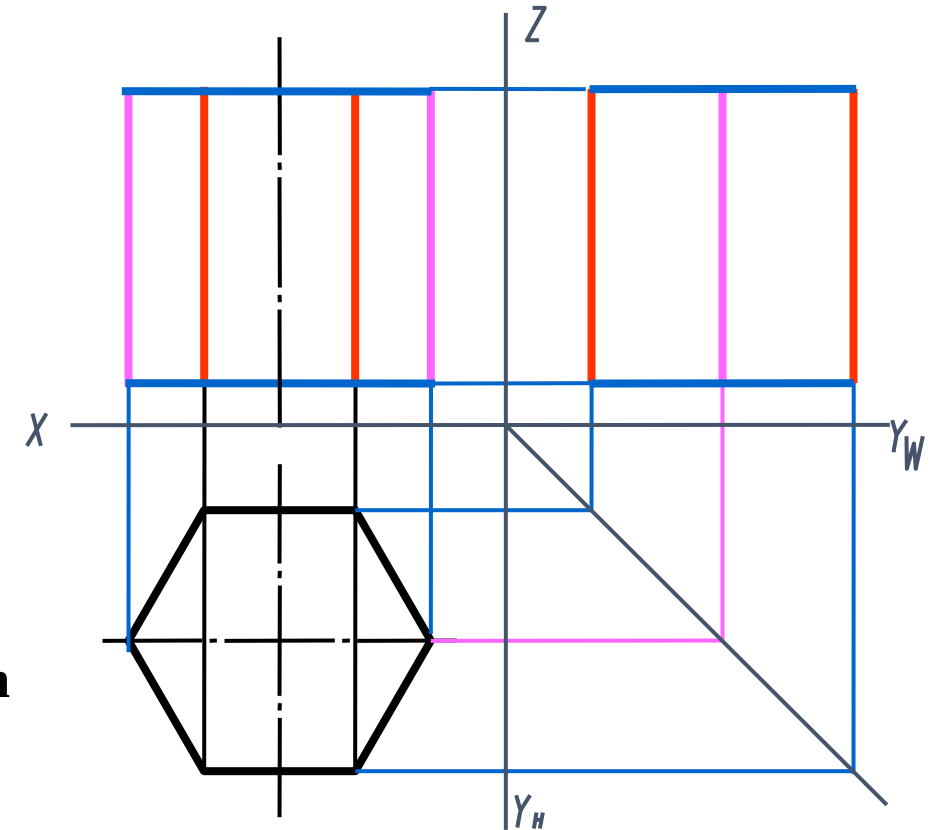
1、 Prism

Drawing Steps:

1. Layout.
2. Start by drawing the top view that reflects the true shape of the bottom face.
3. Next, draw the front view based on the height of the hexagonal prism, **aligning it with the length of the main top view.**
4. Finally, draw the left view, **ensuring the height of the front and left views are aligned, and the width of the top and left views are the same.**
5. Refine and organize the drawing.



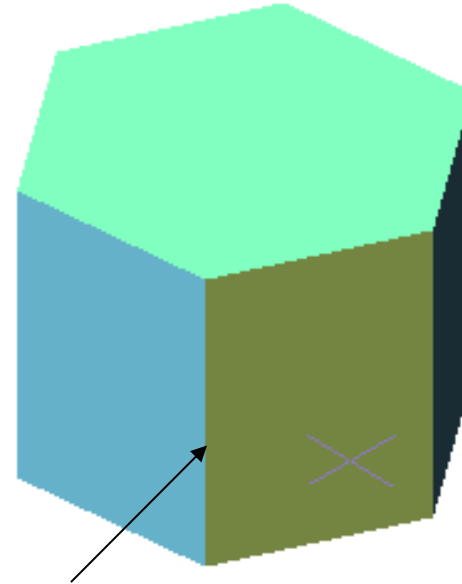
**Front Projection
Direction**



For a prism, simply draw the top and bottom edges. **The endpoints of the edges are the vertices of the bottom polygon.** Connect them accordingly to complete the projection of the prism.

1、Prism

Think



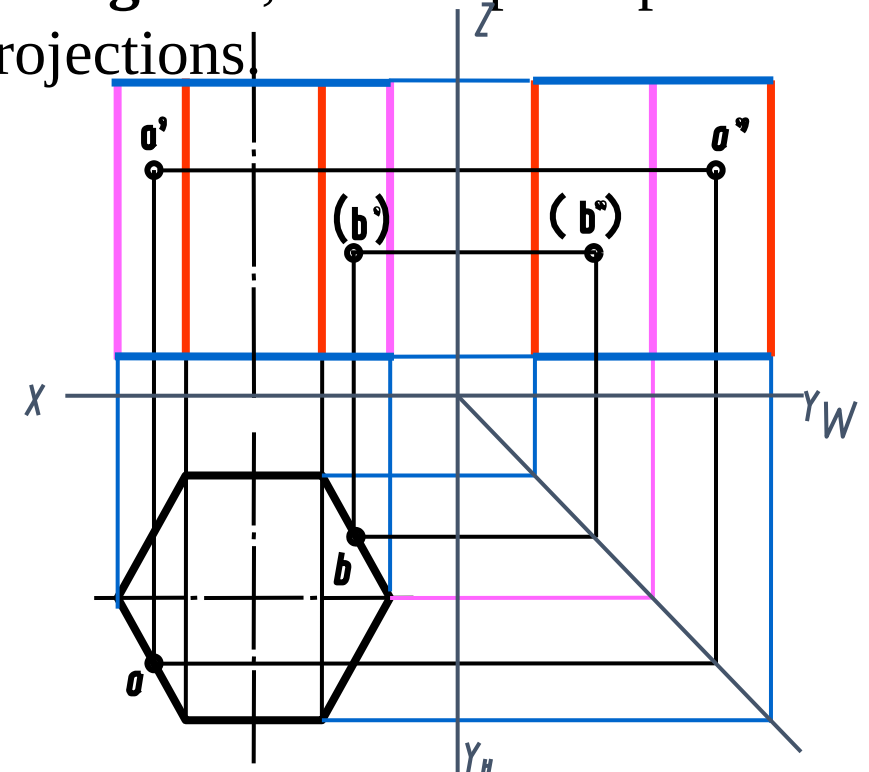
Can the direction directly facing this edge line be the front projection direction?

1、Prism Points on the Surface of a Prism

- (1) First, determine which surface the point lies on.
- (2) If the surface containing the point has **convergence**, directly use the **convergence** to find the remaining projections.
- (3) If the surface containing the point does **not have convergence**, use the principle of **selecting points within a plane** to find the remaining projections!

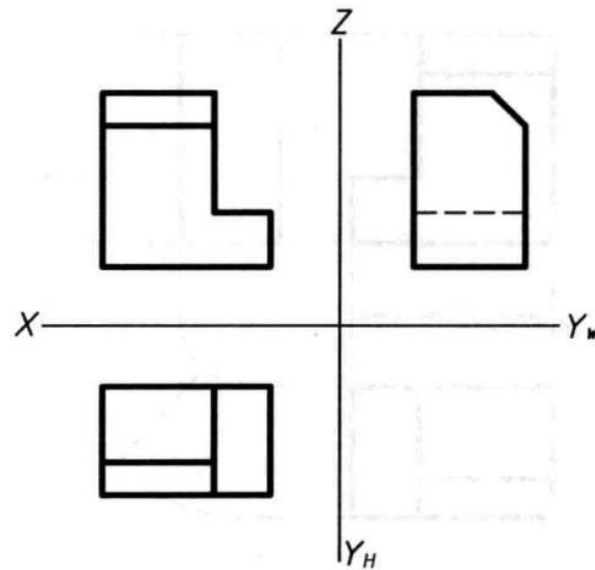
Visibility of Point Projections:

If the projection of the surface containing the point is visible, then the projection of the point on the same surface is also visible; otherwise, it is not visible.



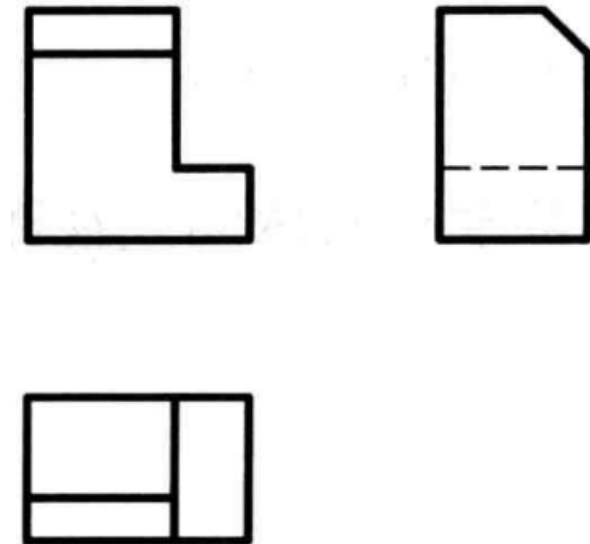
1、Prism

Think: Will the projection axis affect the final three-view projection?



(b)

**Remove the
projection axis.**



(c)

4-1 Three-View Representation of Basic Planar Solids



- The object is placed in different positions within the coordinate system, the coordinates will change, but the shape will not.
- The shape of the object is determined by the relative positional relationship between its elements.

Front Projection $\longrightarrow$ Main View
Reflecting top-bottom, left-right

Three-View System

Remove the
projection axis.

Three Views

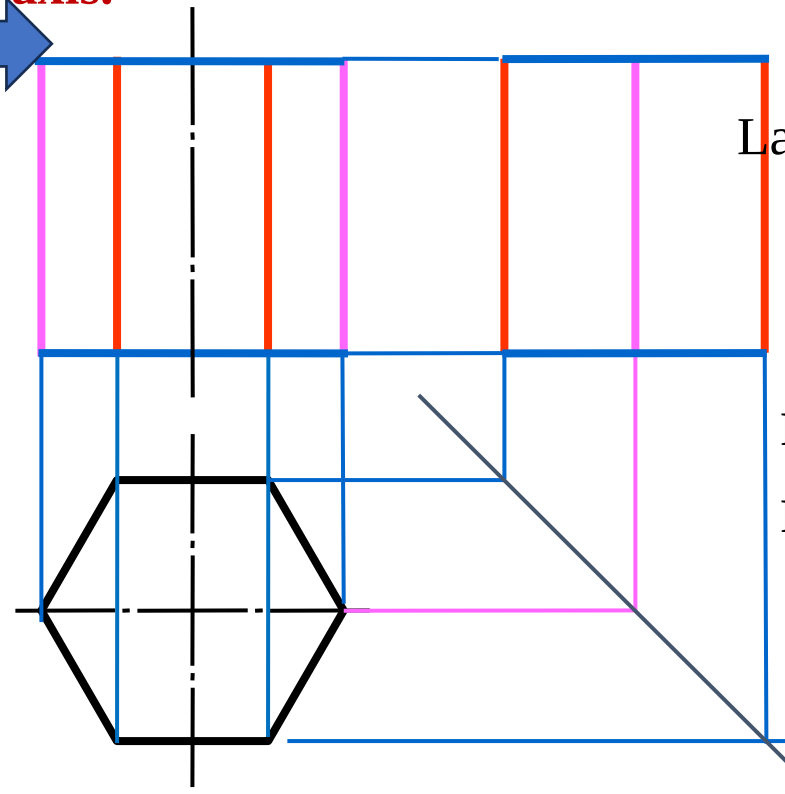
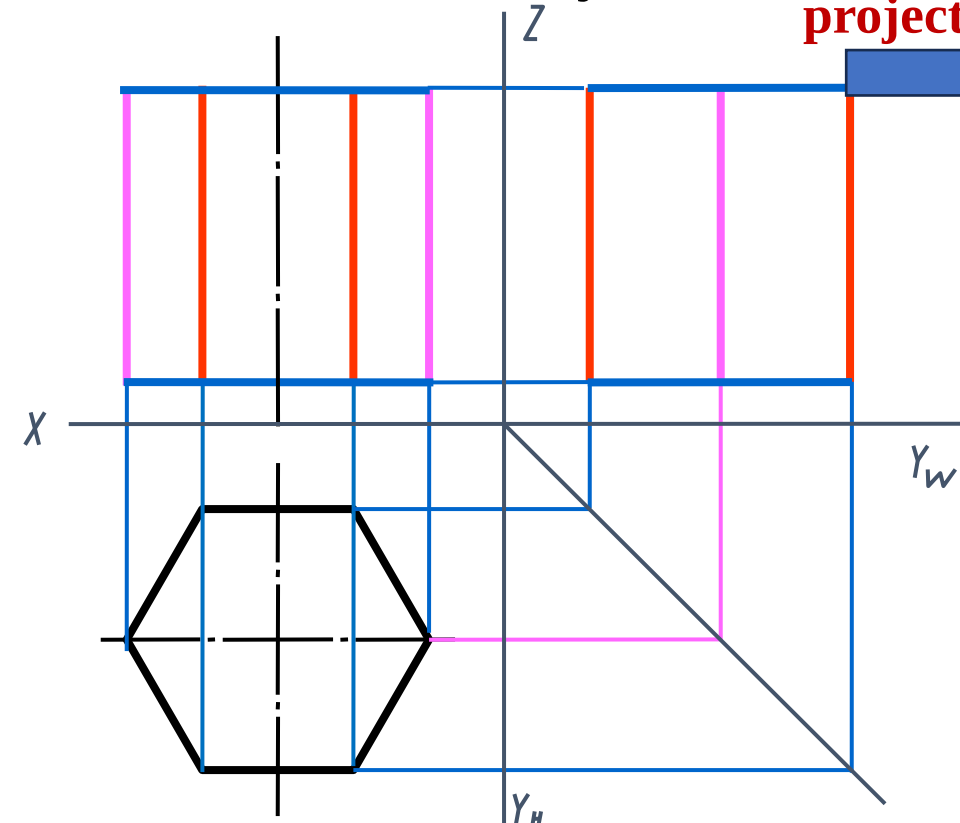
Horizontal Projection $\longrightarrow$ Top View
Reflecting left-right, front-back

Lateral Projection $\longrightarrow$ Left View
Reflecting top-bottom, front-back

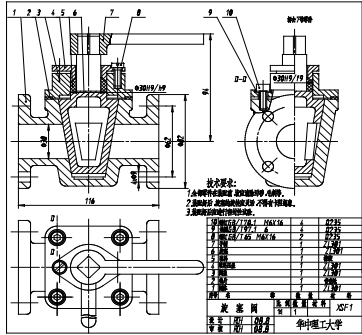
Front and Left — Height is aligned

Front and Top — Length is aligned

Top and Left — Width is equal



4-1 Three-View Representation of Basic Planar Solids



The left view shows the three-view drawing of the assembly.

Think

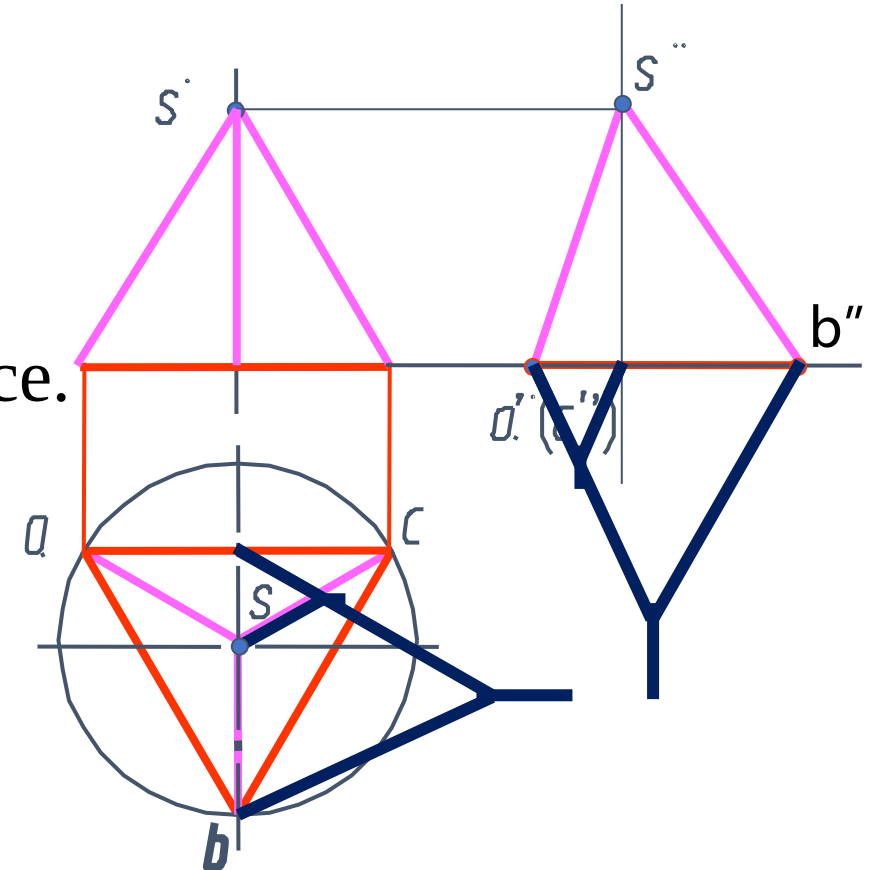
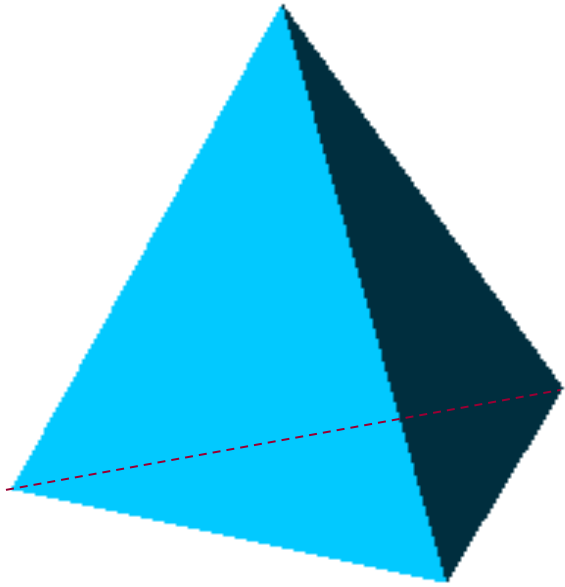
Without a 45° line, how can the Y-axis (width) of the top view and left view be connected?



2、Pyramid Taking a regular triangular pyramid as an example

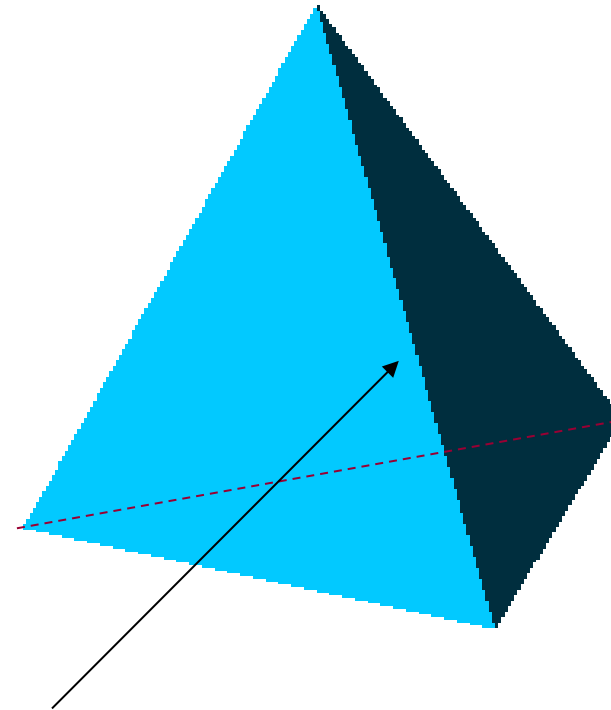
Drawing Steps

- (1) Align properly.
- (2) Analyze each face and determine the drawing reference.
- (3) Draw the base.
- (4) Draw the apex.
- (5) Draw the edges.



2、Pyramid

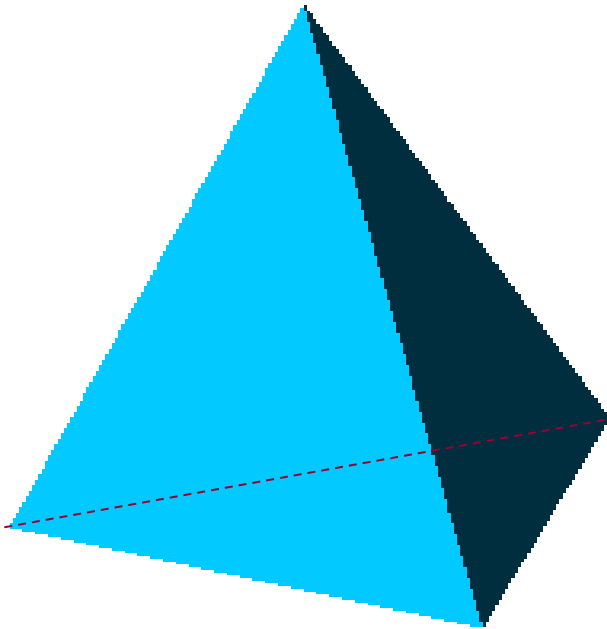
Think



Can the direction directly facing this edge line be the front projection direction?

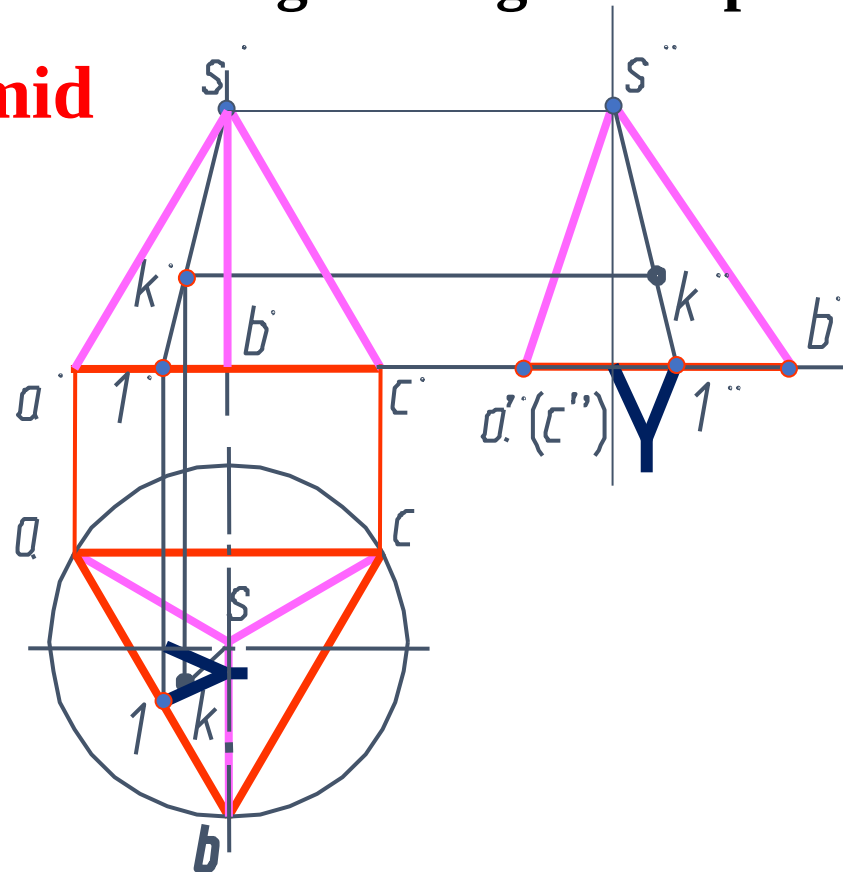
2、Pyramid

Select Points on the Surface of the Pyramid

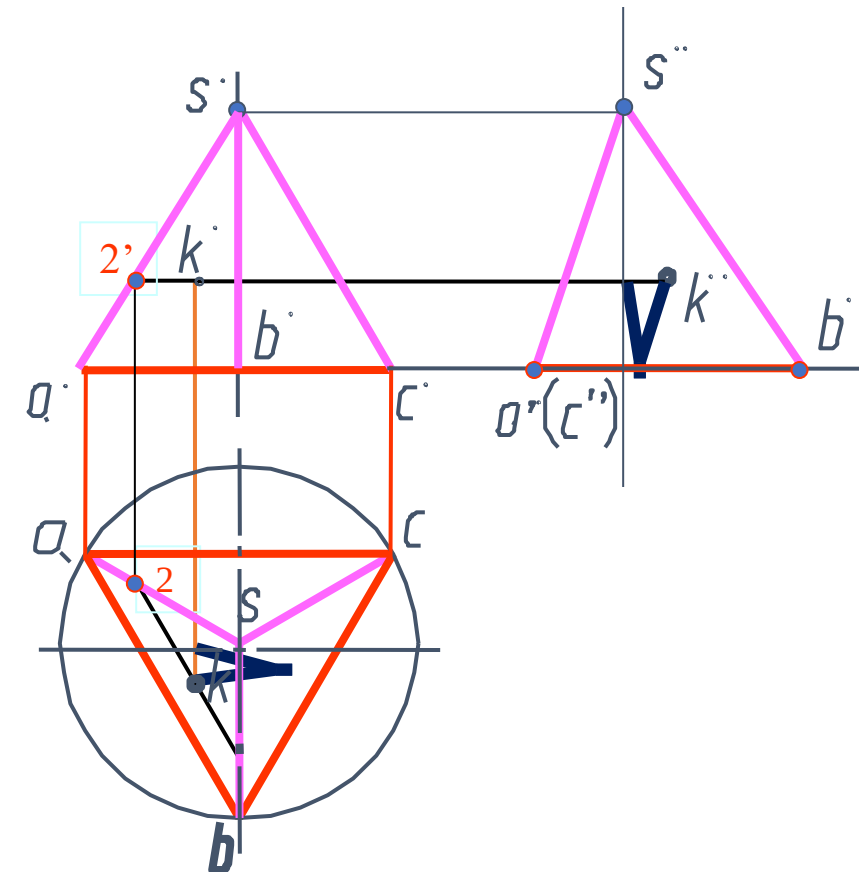


Note: Determine visibility.

Method 1: Draw a Line Passing Through the Apex



Method 2: Draw a Parallel Line



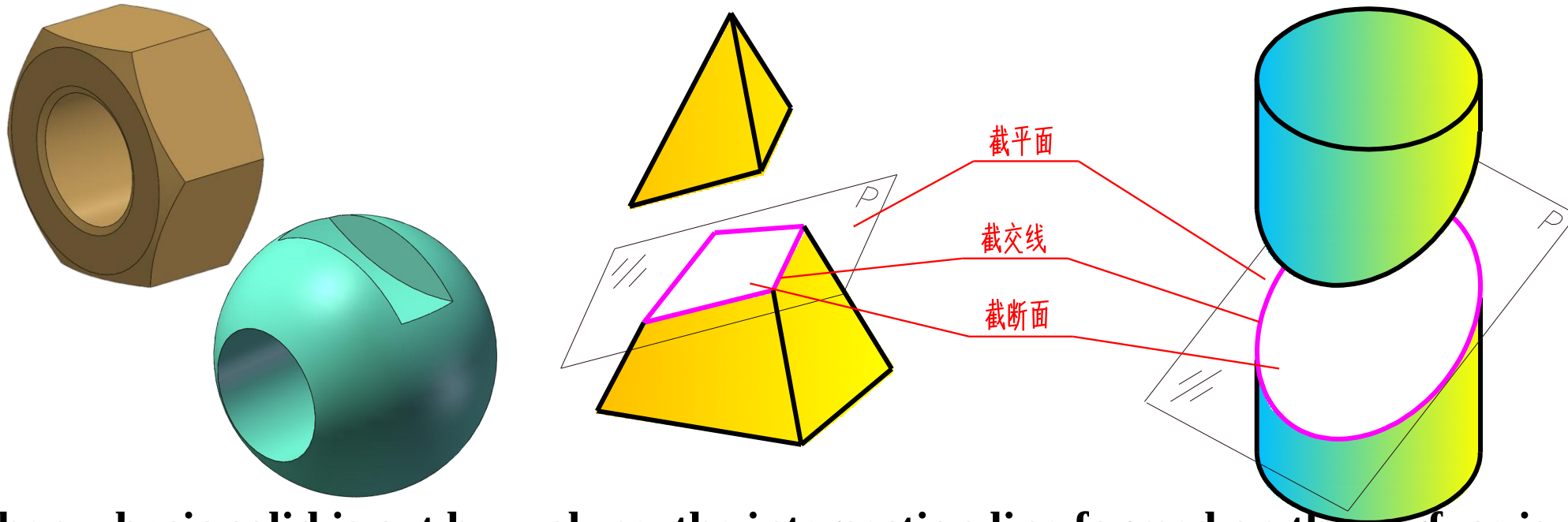
Chapter 4: Basic Bodies and Their Intersection Lines

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4-2 Intersection Lines of Planar Solids



The basic geometric shapes that make up machine parts are often incomplete. For example, the nut and valve core shown in the figure can be regarded as basic solids formed by cutting basic solids with planes.



When a basic solid is cut by a plane, the intersection line formed on the surface is called the **intersection line**. The plane that cuts the solid is called the **cutting plane**. The shape enclosed by the intersection line is called the **cut section**. To accurately represent the shape and structure of the cut solid, the projections of these intersection lines must be drawn.

4-2 Intersection Lines of Planar Solids

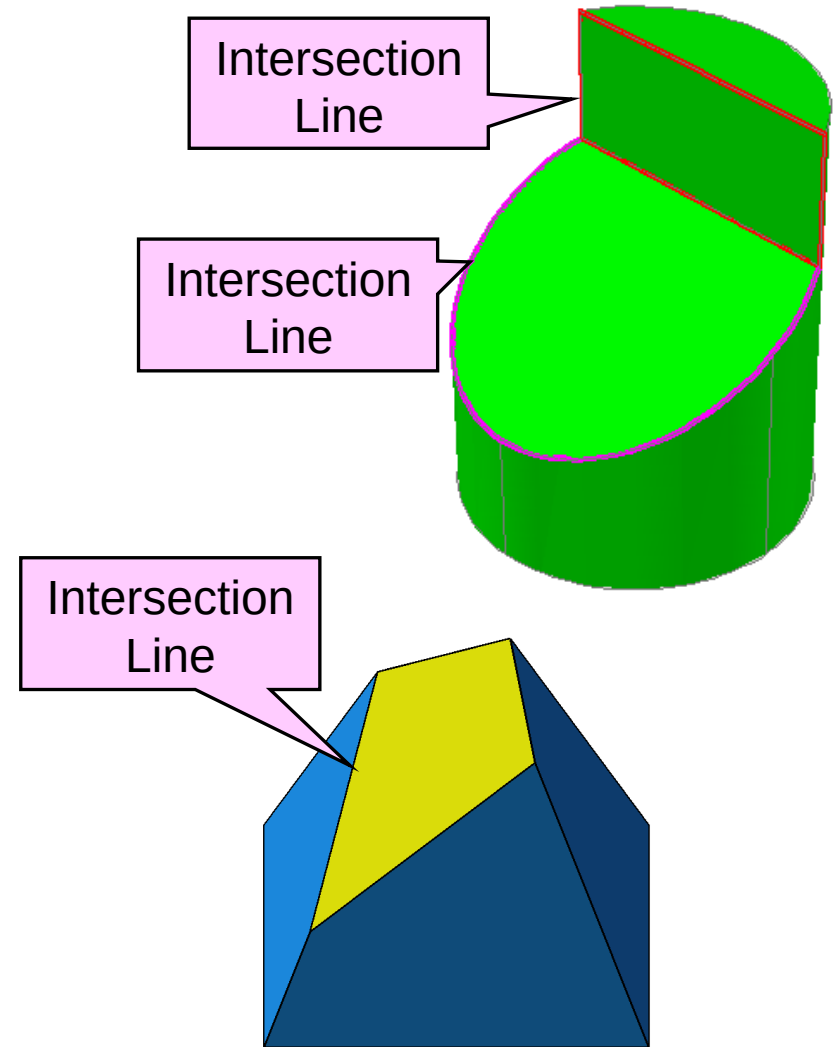


The intersection line has the following properties:

(1) **Commonality**: The intersection line is the shared line between the cutting plane and the surface of the solid. Points on the intersection line lie on both the cutting plane and the solid's surface, making them common points of both.

(2) **Closure**: The intersection line is generally a closed planar shape formed by straight lines, curves, or a combination of both.

——The essence of drawing the intersection line is to find the projections of several **common points** between the cutting plane and the solid's surface.



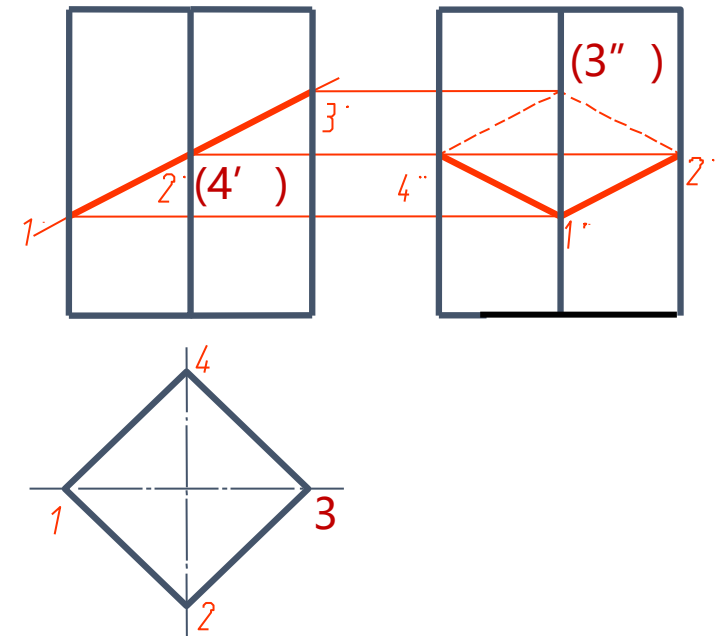
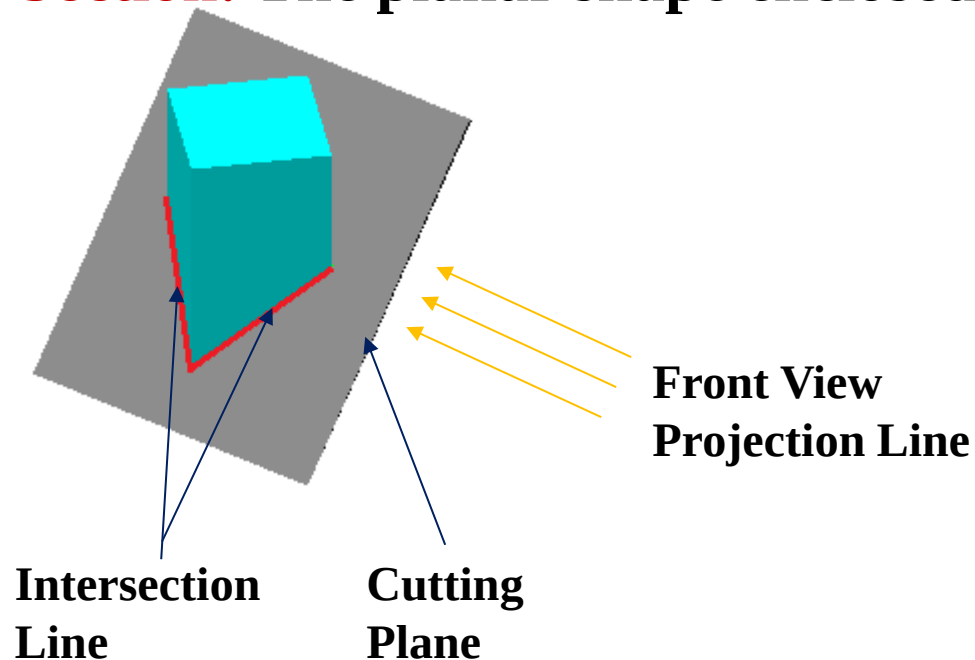
1、 Basic Concepts

If a basic solid is cut by one or more planes, an incomplete basic solid is formed.

Cutting Plane: The plane that cuts the solid.

Intersection Line: The line of intersection between the cutting plane and the surface of the solid.

Cut Section: The planar shape enclosed by the intersection line.



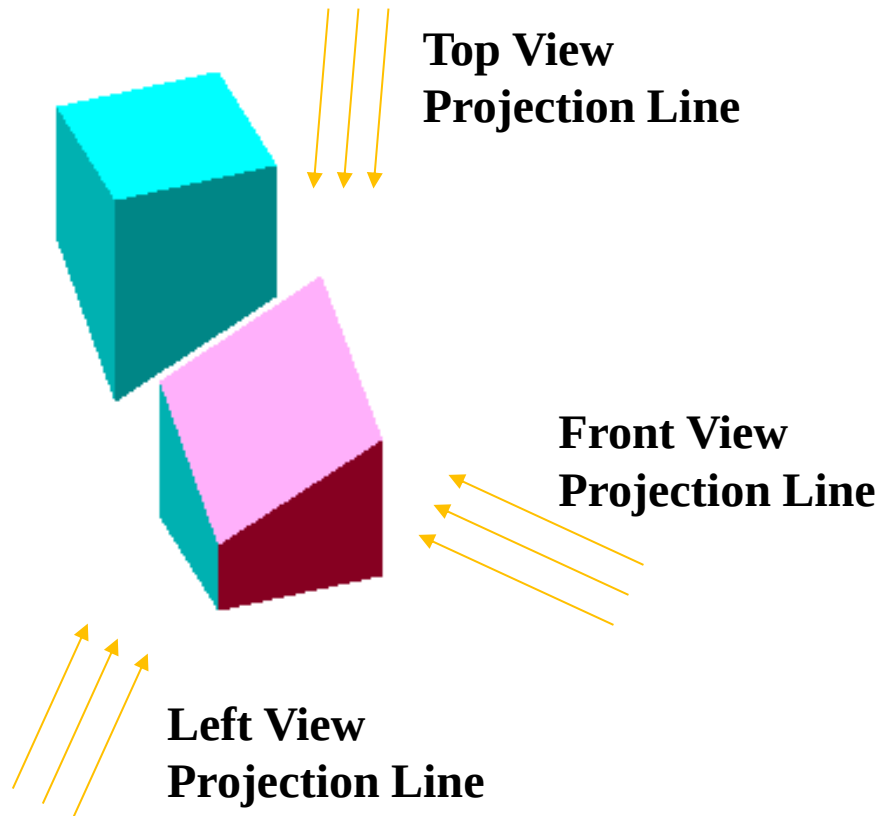
4-2 Intersection Lines of Planar Solids



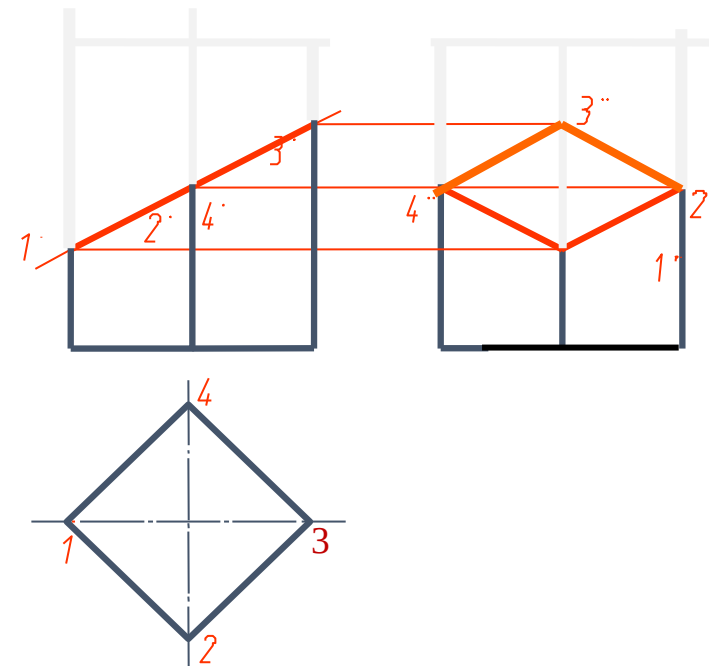
2、Prism with a Cut

Cut by a **Horizontal Plane** or a **Side Vertical Plane**, similarly.

● Prism Cut by a Vertical Plane

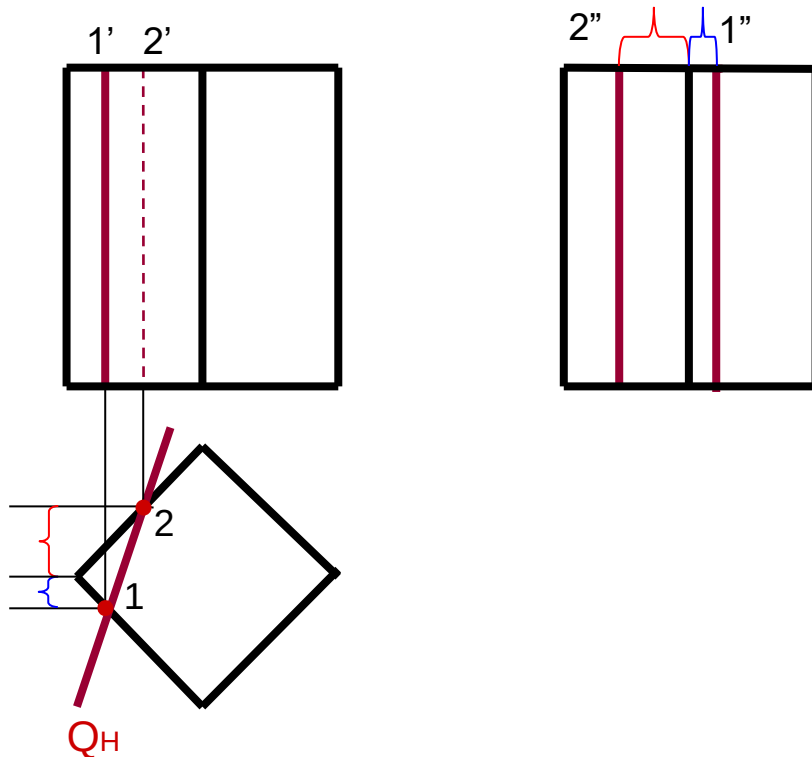


Find the Front Projection of the Intersection Line
Find the Top Projection of the Intersection Line
Find the Side Projection of the Intersection Line



2、Prism with a Cut

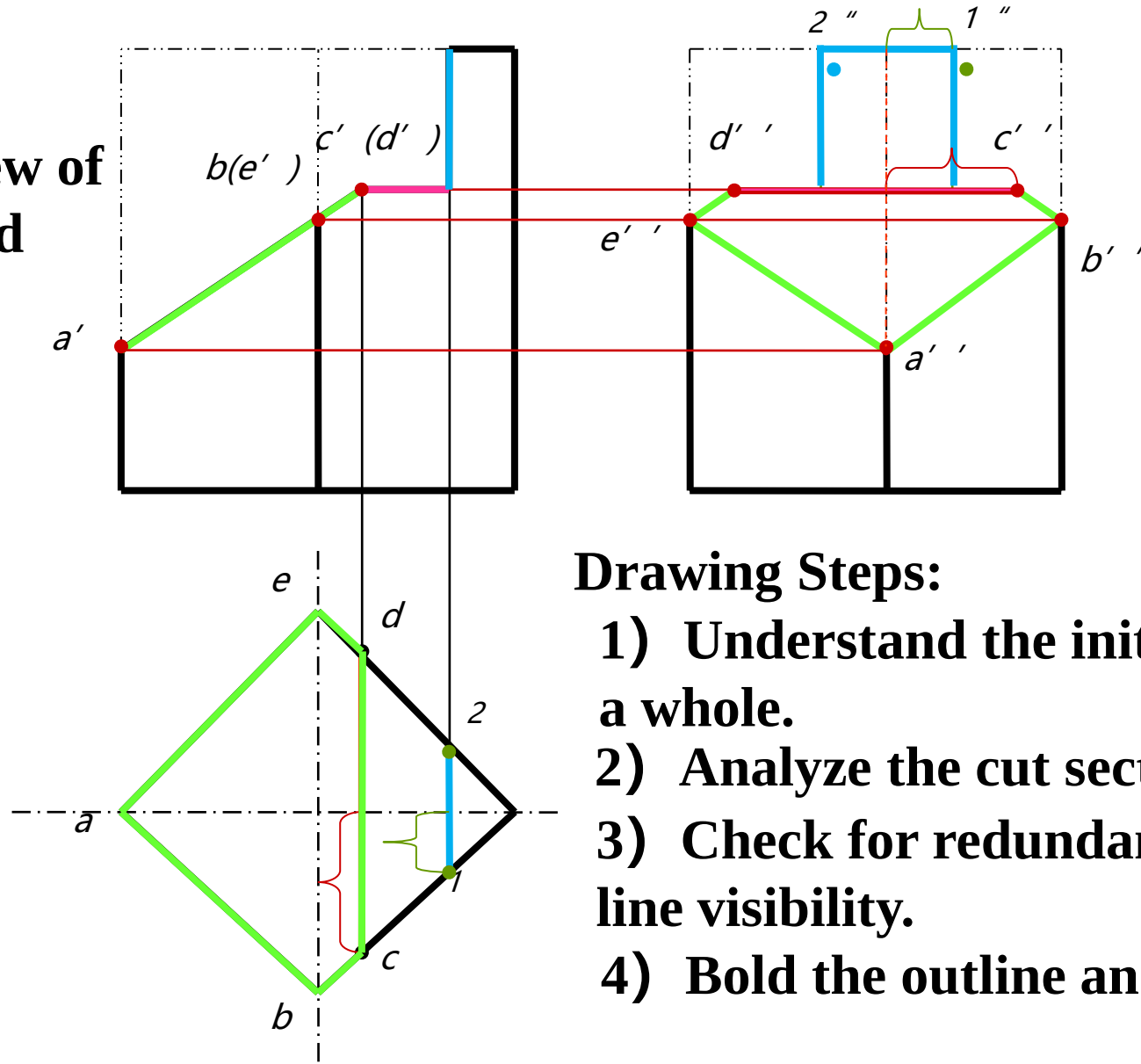
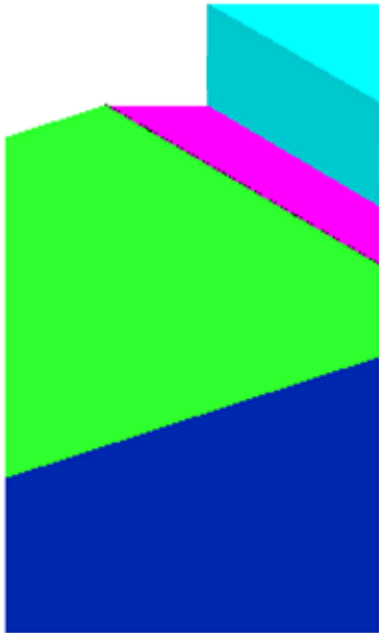
- Prism Cut by a **Vertical Plane**



Cut by a **Frontal Plane** or a **Side Plane**, similarly.

2、Prism with a Cut

Example: Given the front view of a prism, find the top view and the left view.



Drawing Steps:

- 1) Understand the initial shape as a whole.
- 2) Analyze the cut sections.
- 3) Check for redundant lines and line visibility.
- 4) Bold the outline and main lines.

2、Prism with a Cut

See the exercise questions.

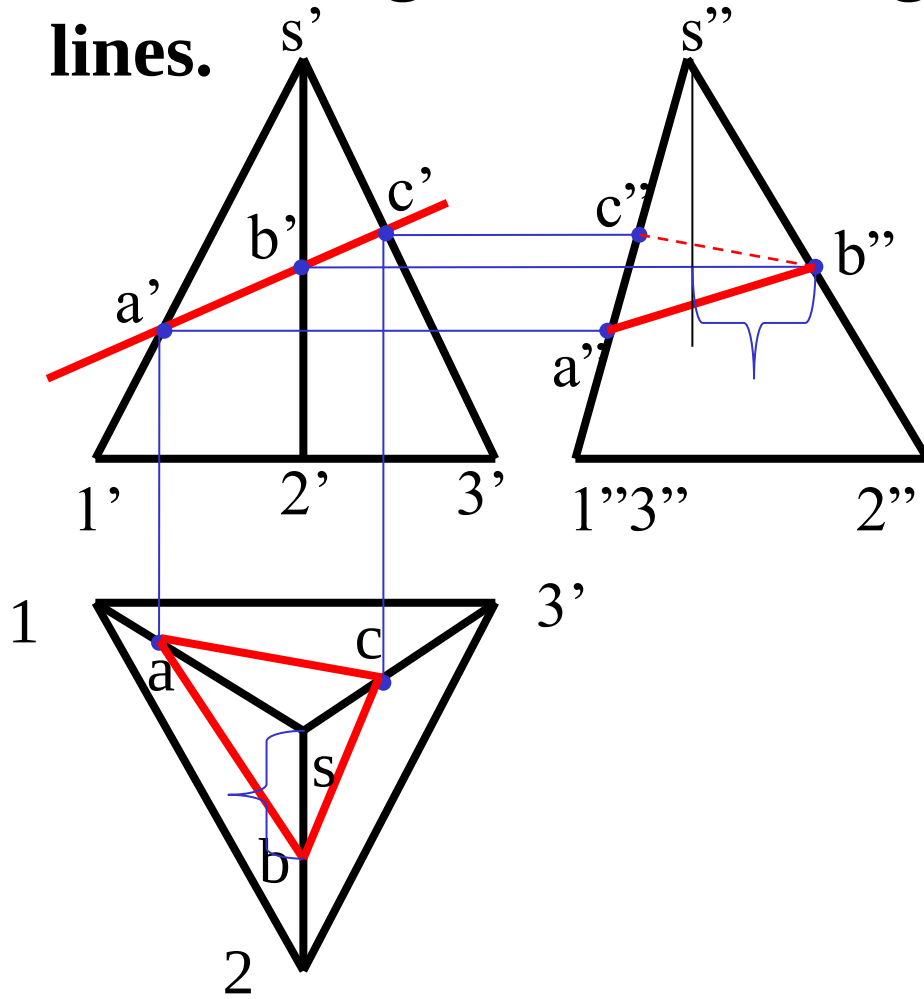
**Students, let's practice together ✎,
and I will teach you step by step until you master it! 😊**

4-2 Intersection Lines of Planar Solids

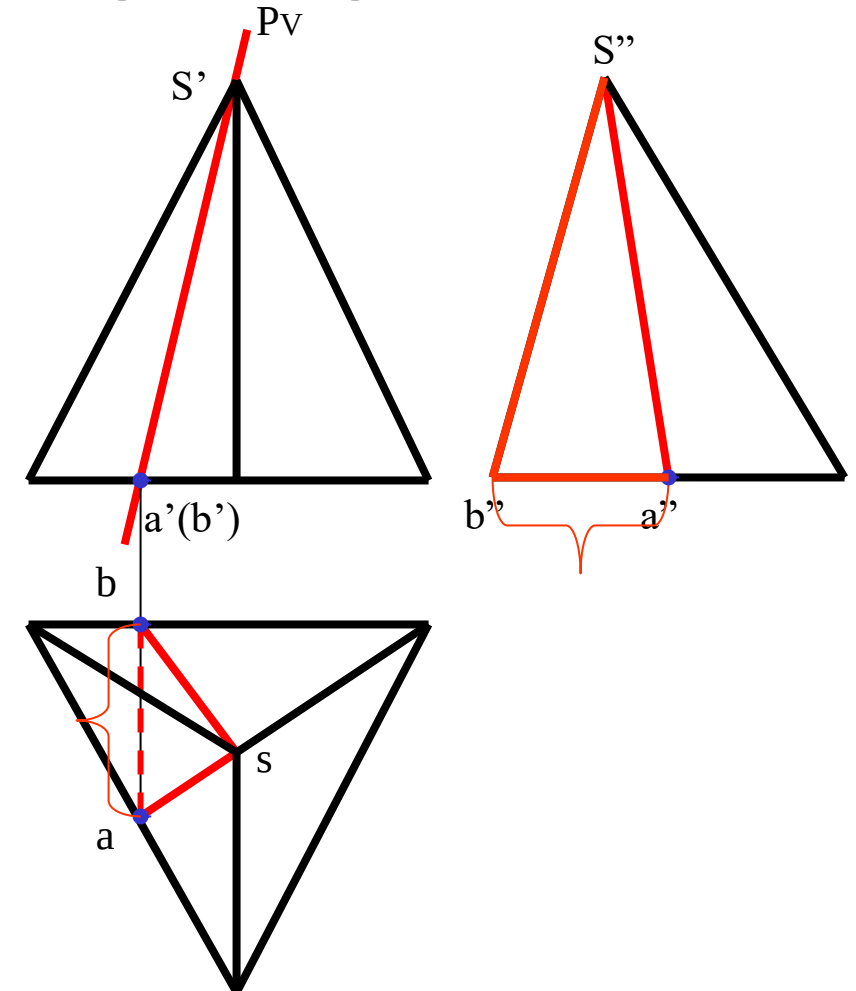


3、Pyramid with a Cut

- A prism cut by a **vertical plane**, intersecting with all the edge lines.



- A prism cut by a **vertical plane**, passing through the apex.

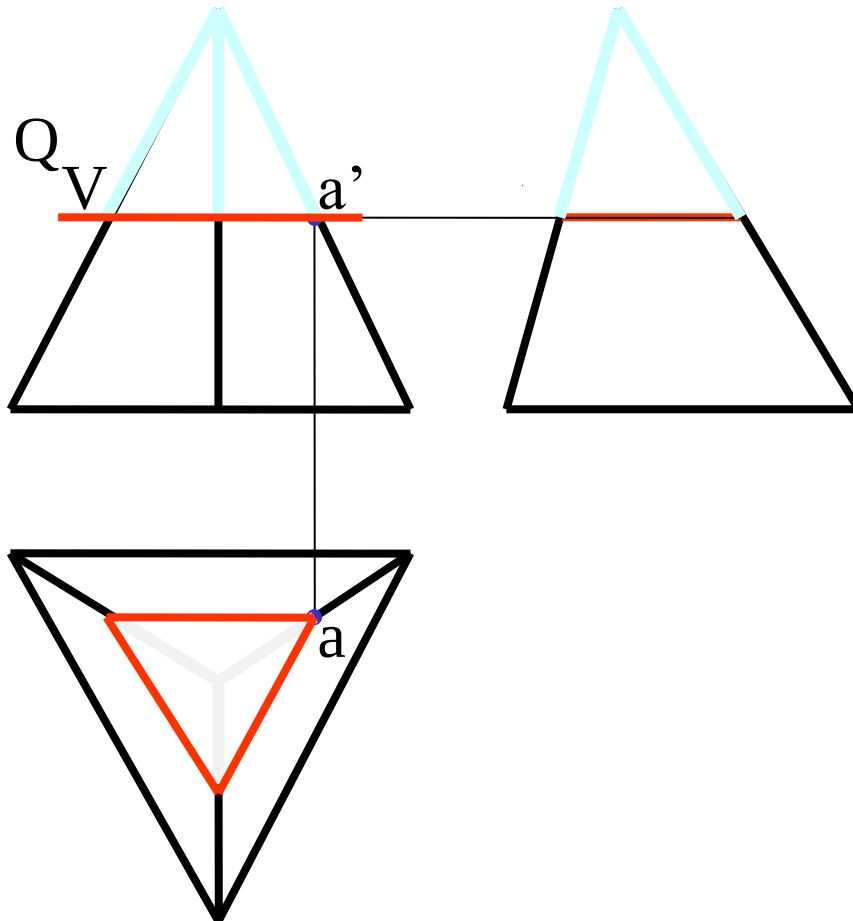


4-2 Intersection Lines of Planar Solids

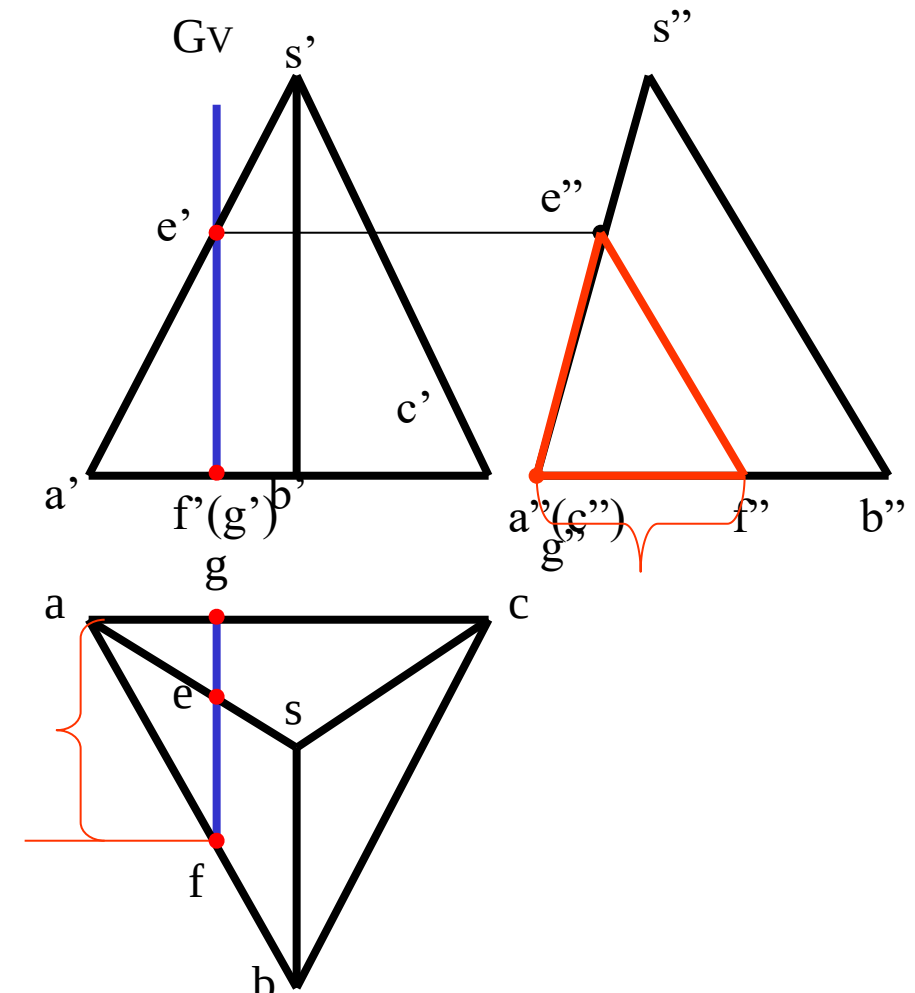


3、Pyramid with a Cut

- A prism cut by a **horizontal plane**.



- A prism cut by a **side plane**.

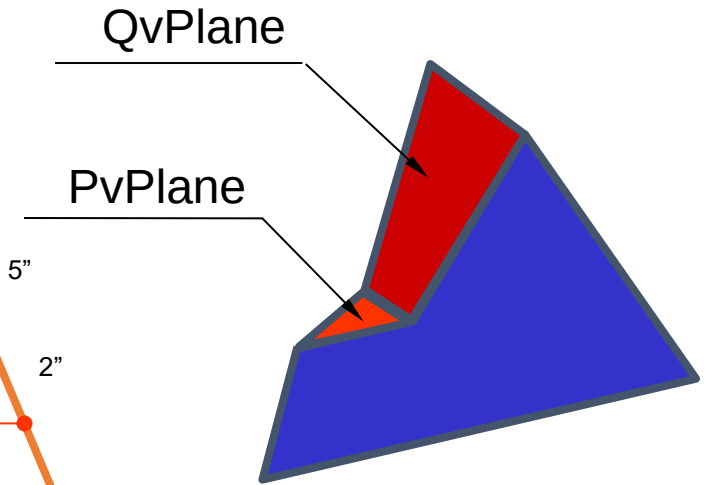
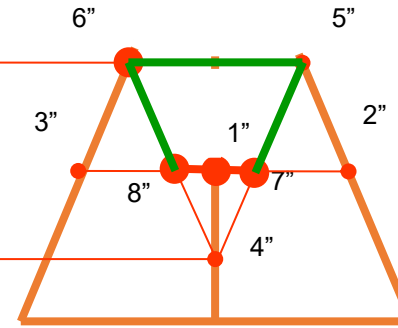
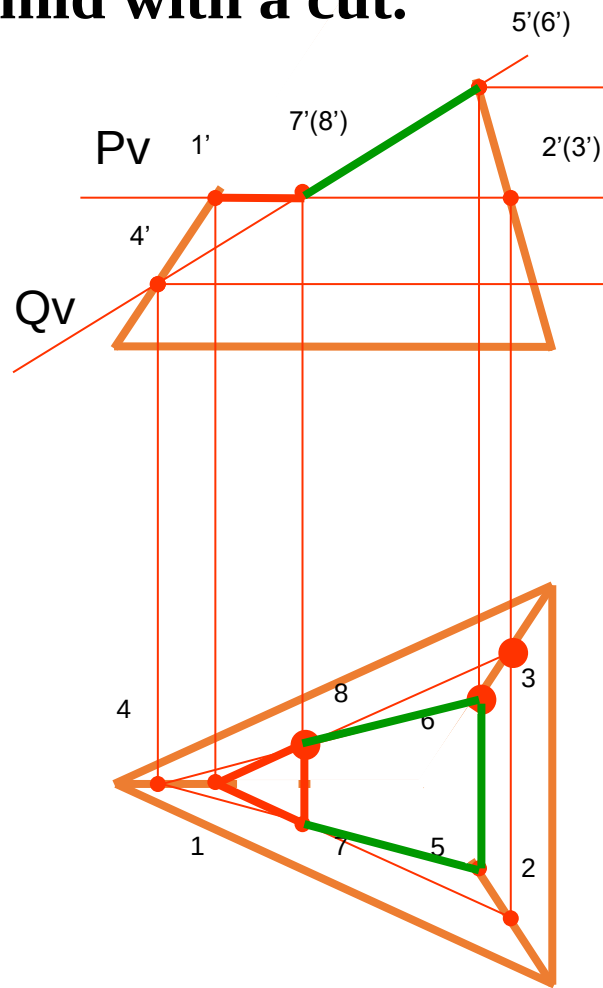


4-2 Intersection Lines of Planar Solids



3、Pyramid with a Cut

Example: Draw the three views of a triangular pyramid with a cut.



1.Cut: Formed by cutting a part of the solid with one or more planes.

2.Key: Draw the projections of the intersection lines between the plane and the surface of the solid.

3.Steps: First, consider the whole object, then the cut. Start with the full cut, then proceed to the local cut.

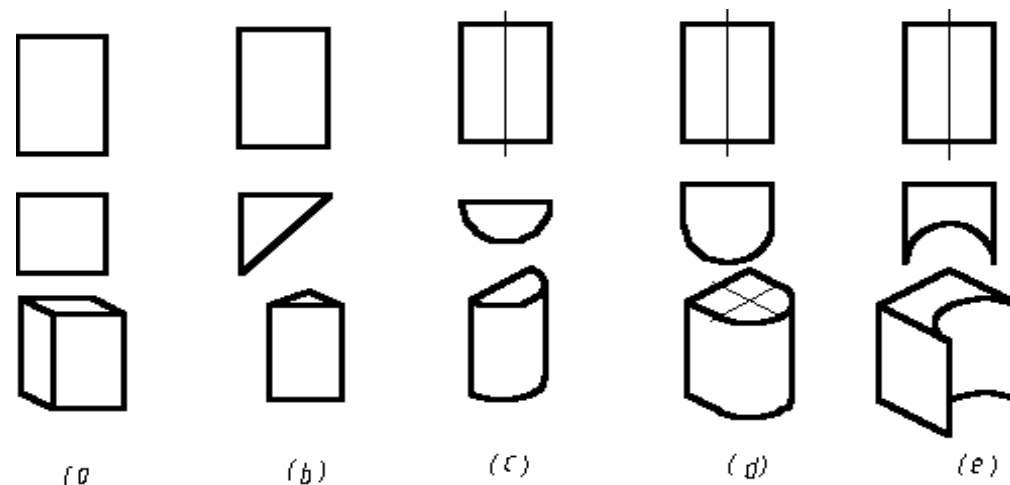
3、Pyramid with a Cut

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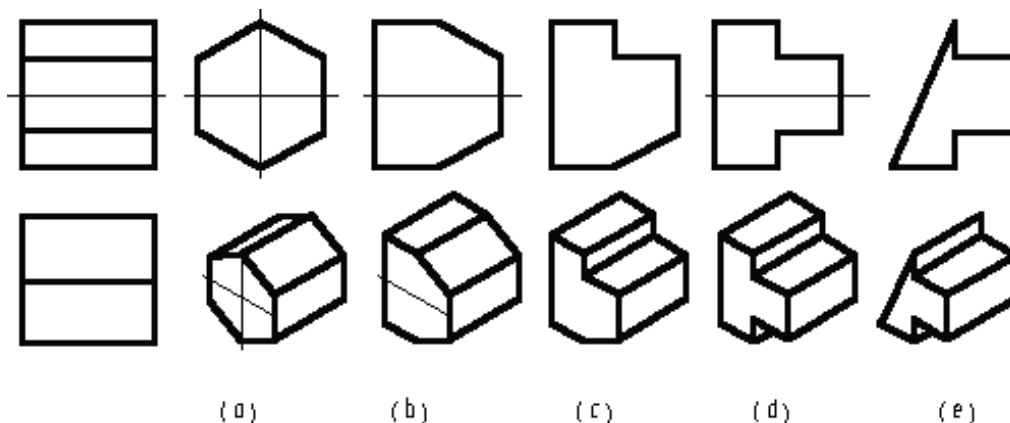
**Students, let's practice together ✎,
and I will teach you step by step until you master it! 😊**

4、 Reading Method — Line and Plane Analysis Method

- One view cannot determine the shape of an object. For example, the front view may be the same, but it cannot determine its spatial shape.



- Two views also cannot determine the shape of an object. For example, if the front and top views are the same, but the left view is different, the spatial shape cannot be determined.



Only by **combining the three views** can the spatial shape of the object be determined.

4、 Reading Method — Line and Plane Analysis Method

Definition: A method of analyzing the surface properties and relative positions of an object using the **lines and wireframes** in the views to **imagine the spatial shape of the object**.

- Projection is a geometric element of a **straight line**.

It can be a **straight line** (edge line, generatrix line (contour line of rotation), intersection line), or it can be **the projection of the plane's convergence**.

- Projection is a geometric element of a closed line frame.

In general, a closed line frame in a view represents **the projection of a surface of the object**. Different line frames represent different surfaces, which can be planes, curved surfaces, or composite surfaces formed by the intersection of **planes and curved surfaces or curved surfaces with each other**.

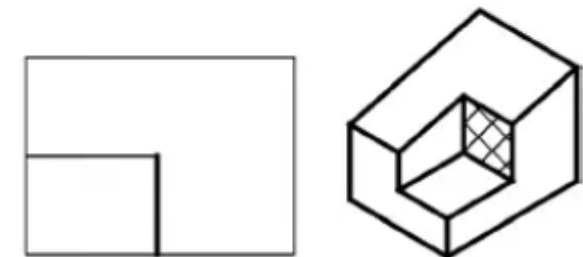
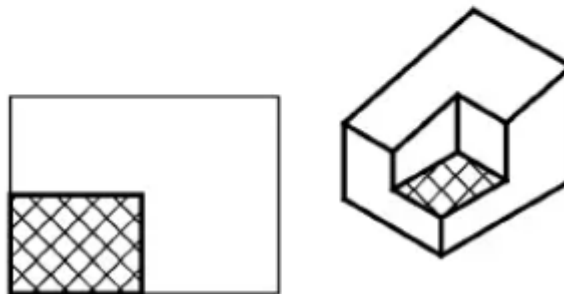
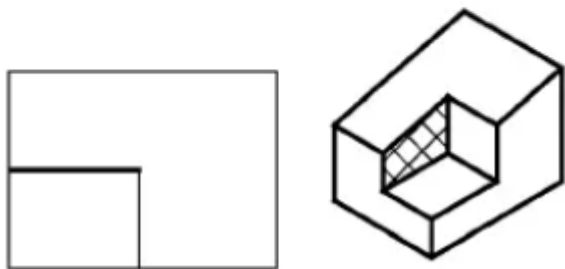
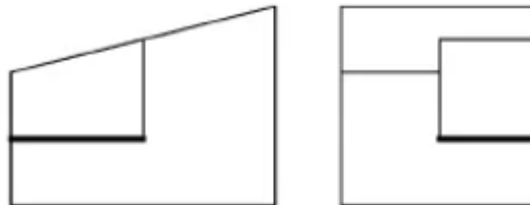
4-2 Intersection Lines of Planar Solids



4、 Reading Method — Line and Plane Analysis Method

Definition: A method of analyzing the surface properties and relative positions of an object using **the lines and wireframes** in the views to **imagine the spatial shape of the object**.

"Length alignment, height alignment, and equal width" projection rule for the three views, **involves line frames** and finding corresponding projections.



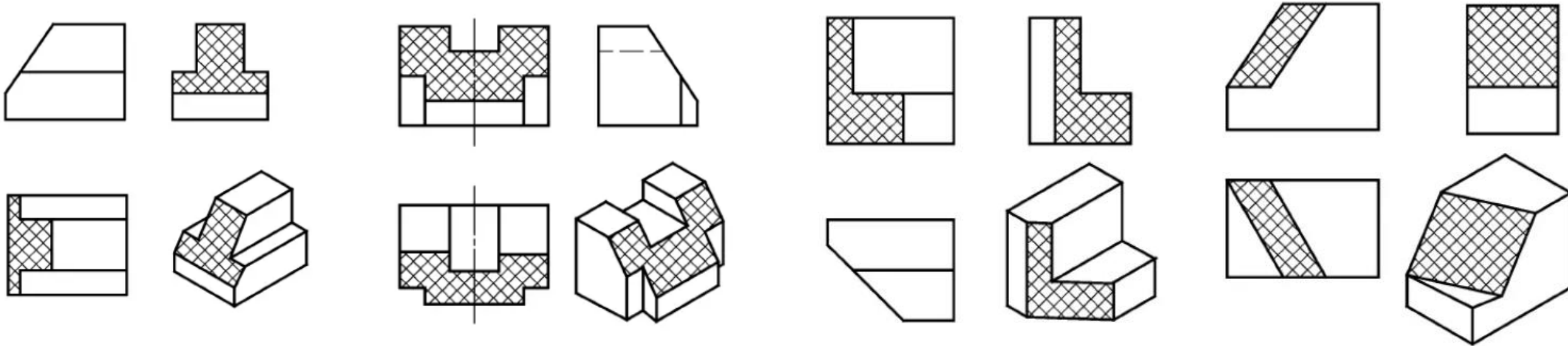
Frontal Plane

Horizontal Plane

Side Plane

4、Reading Method — Line and Plane Analysis Method

"Length alignment, height alignment, and equal width" projection rule for the three views, **involves line frames and finding corresponding projections.**



Frontal Vertical
Plane

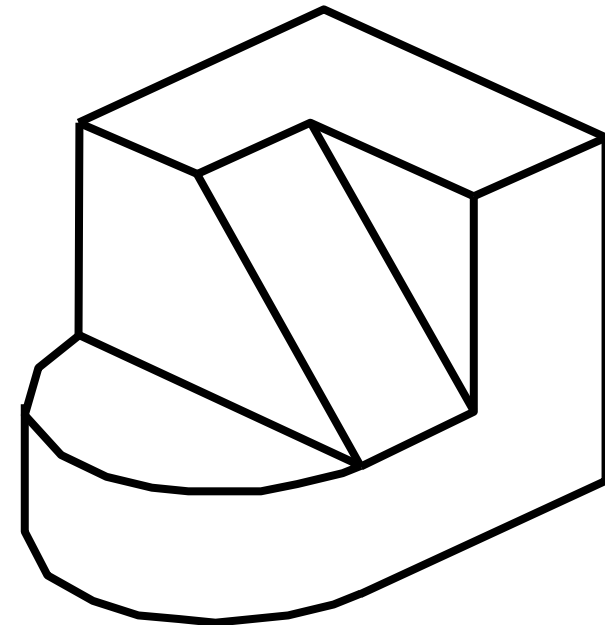
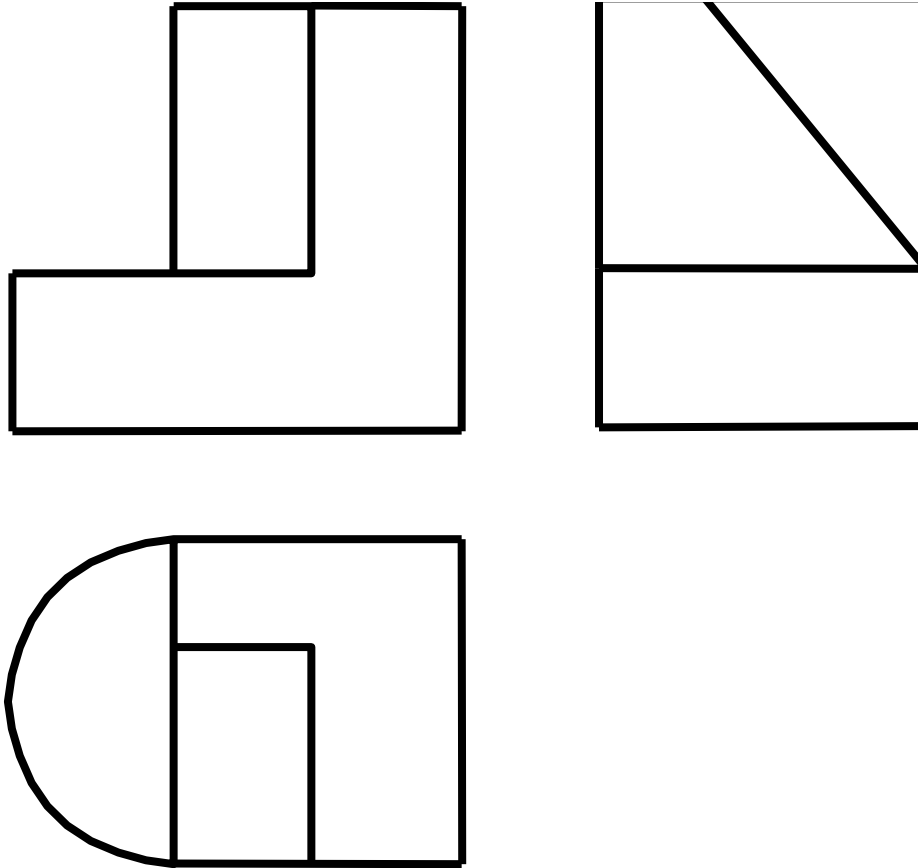
Side Vertical
Plane

Vertical Plane

General Plane

4、 Reading Method — Line and Plane Analysis Method

"Length alignment, height alignment, and equal width" projection rule for the three views, **involves line frames and finding corresponding projections.**



4、 Reading Method — Line and Plane Analysis Method

Example: Given the front view and left view of a solid, find the top view.

